



BAHRIA UNIVERSITY, (Karachi Campus)

Department of Software Engineering

PROJECT PROPOSAL

Course Title: Software Application for Mobile Devices **Course Code:** SEL-448
Course Instructor: Engr. Adnan-ur-Rehman **Class:** BSE-6(A)
Lab Instructor: Engr. Saniya Sarim **Name:** _____
Date: 4-17-2026

PROJECT TITLE:



Awwab

GROUP MEMBERS LIST:

<WITH TEAM LEAD>

Name	Enroll
Maria Khan (Lead)	02-131232-114
Muhammad Umer Qureshi	02-131232-108
Muhammad Umer	02-131232-090

PROJECT SCOPE:

The scope of this project is to design and develop a cross-platform mobile application that assists users in offering prayers on time through intelligent reminders and accurate prayer timing calculations. The application will utilize location-based services to dynamically fetch prayer timings and provide timely Azan alerts.

The system will be developed using Flutter for mobile application development and integrated with cloud computing services (Microsoft Azure) for backend processing, data storage, and scalability. The application will support both online and offline functionality, ensuring uninterrupted usability.

The project includes key features such as prayer time notifications, manual reminders, Namaz guidance, Qibla direction, customizable Azan sounds, daily Hadith notifications, and multi-language support (English & Urdu).

However, features such as user authentication, social integration, and wearable device support are considered out of scope for the current version and may be implemented as future enhancements.

PROJECT ABSTRACT:

In the modern digital era, maintaining consistency in daily prayers can be challenging due to busy schedules, environmental distractions, and lack of accessible prayer reminders. This project presents the development of a Smart Namaz Companion mobile application that aims to assist users in managing their prayer routines effectively.

The application leverages location-based services to provide accurate prayer timings and generates automated Azan alerts. Additionally, it offers customizable reminders, offline functionality, Qibla direction, and a built-in Namaz guidance module to support users in performing prayers correctly.

Furthermore, the application enhances user engagement by providing daily Hadith notifications with translation and brief explanation, along with multi-language support (English and Urdu) to improve accessibility.

The system integrates mobile application development using Flutter with cloud-based services (Microsoft Azure) for efficient data handling, scalability, and performance optimization. By combining real-time notifications, user-friendly design, and reliable backend support, the application provides a comprehensive solution for enhancing religious practice through technology.

PROJECT FUNCTIONALITIES:

1. Location-Based Prayer Timing

The application will automatically detect the user's location and fetch accurate prayer timings using external APIs. It will dynamically adjust timings based on geographical location and seasonal variations.

2. Automated Azan Alerts

The system will generate notifications at the exact time of each prayer, accompanied by authentic Azan audio to alert the user.

3. Custom Prayer Reminders

Users will be able to set additional reminders (e.g., 5–10 minutes after Azan) to ensure they do not forget to perform نماز.

4. Namaz Guidance Module

The application will include a structured guide explaining the number of Rakats, steps, and method of performing each prayer.

5. Qibla Direction (Compass Integration)

The system will provide real-time Qibla direction using device sensors and location data.

6. Offline Functionality

The application will store previously fetched prayer timings locally, allowing it to function without internet connectivity.

7. Dark Mode Support

A theme-switching feature will be implemented to enhance usability in low-light conditions.

8. Azan Sound Customization

Users will be able to select from a set of preloaded authentic Azan sounds according to their preference.

9. Daily Hadith Notification System

The application will send one Hadith daily as a notification, including translation (Tarjuma) and brief explanation.

10. Multi-Language Support

The application will support English and Urdu, allowing users to switch language through settings.

11. Cloud Integration (Microsoft Azure)

The application will utilize cloud services for storing user preferences, managing Hadith data, and ensuring scalability and reliability.

1. Location-Based Prayer Timing

Automatically detects user location and fetches accurate prayer timings using APIs with seasonal adjustments.

2. Automated Azan Alerts

Triggers notifications at each prayer time with optional Azan audio playback.

3. Custom Prayer Reminders

Allows users to set extra reminders (e.g., 5–10 minutes before/after prayer time).

4. Namaz Guidance Module

Provides step-by-step guidance for all prayers including Rakats, posture, and method.

5. Qibla Direction (Compass Integration)

Uses GPS and device sensors to show real-time Qibla direction accurately.

6. Offline Functionality

Stores prayer times and user preferences locally for uninterrupted usage without internet.

7. Dark Mode Support

Provides theme switching for better usability in different lighting conditions.

8. Azan Sound Customization

Allows users to choose different Azan voices for different prayer times.

9. Daily Hadith Notification System

Sends one Hadith daily with translation and short explanation.

10. Multi-Language Support

Supports English and Urdu for wider accessibility.

11. Cloud Integration (Microsoft Azure)

Used for storing Hadith data, syncing preferences, and ensuring scalability.

12. Prayer Consistency Tracker

Tracks user prayer habits and builds a consistency record:

- Daily/weekly/monthly prayer completion tracking
- Streak system for motivation
- Calendar-based visualization of prayer activity

13. Smart Silent Mode Detection

Detects phone sound mode and ensures Azan notifications are still effective:

- Overrides silent mode (optional setting)
- Uses vibration fallback if audio is disabled
- Smart notification handling based on user preference

14. Mosque Finder Nearby

Uses GPS to locate nearby mosques:

- Shows distance and directions
- Integrates with maps for navigation
- Helps users find prayer locations easily while traveling

15. Tasbeeh Counter (Digital Counter)

A built-in digital dhikr counter:

- Tap-based counting system
- Reset and session tracking
- Optional saved counters for daily zikr goals

MODULE DISTRIBUTION:**1. Location & Prayer Time Module**

This module handles fetching the user's current location and calculating accurate prayer timings using APIs. It adjusts automatically based on city, timezone, and seasonal changes.

2. Azan Notification System

Responsible for triggering automatic prayer alerts at exact times. It also plays Azan audio and manages notification scheduling even when the app is in background.

3. Qibla Compass Module

Uses device sensors (magnetometer + GPS) to show real-time Qibla direction. Ensures accuracy regardless of user location.

4. Namaz Guidance Module

Provides step-by-step guidance for all prayers, including Rakats, actions, and instructions in a simple UI for beginners.

5. Offline Storage Module

Stores previously fetched prayer times and user preferences locally so the app works without internet connectivity.

6. UI/UX & Theme Module

Handles overall design, including dark mode, language switching UI, and smooth navigation between screens.

7. Hadith Notification Module

Fetches and displays one Hadith daily with translation and short explanation through push notifications.

8. Cloud Integration Module (Azure)

Manages backend services like storing Hadith data, user preferences, and ensuring scalability and synchronization.

9. Prayer Consistency Tracker Module

Tracks user prayer activity and builds a consistency record:

- Daily, weekly, and monthly tracking
 - Prayer streak system
- Visual calendar-based progress

10. Smart Silent Mode Detection Module

Detects device sound mode and ensures prayer notifications are still effective:

- Optional override of silent mode
 - Vibration fallback support
- Smart notification handling

11. Mosque Finder Nearby Module

Uses GPS to locate nearby mosques:

- Shows nearest mosques based on user location
 - Distance and navigation support
 - Integration with map services

12. Tasbeeh Counter (Digital Counter) Module

A built-in digital dhikr counter:

- Tap-based counting system
- Reset and session tracking
- Optional saved zikr goals

Modules	Maria Khan (Lead)	Muhammad Umer Qureshi	Muhammad Umer
Location & Prayer Time Module	✓		
Azan Notification System	✓	✓	
Qibla Compass Module		✓	✓
Namaz Guidance Module	✓		✓
Offline Storage Module		✓	✓
UI/UX & Theme Module	✓	✓	
Hadith Notification Module	✓		✓
Cloud Integration (Azure)	✓	✓	
Prayer Consistency Tracker	✓	✓	
Smart Silent Mode Detection		✓	✓
Mosque Finder Nearby	✓	✓	
Tasbeeh Counter (Digital Counter)		✓	✓
Testing & Debugging	✓	✓	✓
Documentation & Report Writing	✓		✓

Teacher Signature: _____

Remarks: _____

Submission Date: _____